The empirical recurrence for OEIS A235845: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

7 September 2026

Abstract

OEIS A235845 records "Empirical recurrence of order 55", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 72$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 55, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 55 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A235845 is "Number of (n+1)X(3+1) 0..3 arrays with the upper median unequal to the lower median in every 2X2 subblock.". It has offset 1 and begins

$$16228, 1133608, 80556656, 5778908484, 415698469568, 29937676955084, 2156857770093828, 155414194184061336$$

The entry states:

> Empirical recurrence of order 55 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 08:53:09, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 72$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 72$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 144$ exact terms of the model returns order 55 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{55} c_i \, a(n-i),$$

c_1	162	c_{15}	−4065421498811276325	c_{29}	691145490852581833647371 3262
c_2	−8664	c_{16}	−48185133270301590446	c_{30}	11636845106989384763563752150
c_3	126383	c_{17}	261468318691032394094	c_{31}	−35426533241460142847913017292
c_4	4311748	c_{18}	1671049089069060736316	c_{32}	−65194615966312136351702370731
c_5	−166780203	c_{19}	−10731734966440442081791	c_{33}	119511263554884279380939049301
c_6	680304883	c_{20}	−42663948745824865695020	c_{34}	255468111089214931399366366807
c_7	46054839213	c_{21}	303834378738297560532393	c_{35}	−241581567345818140025993633534
c_8	−579812585572	c_{22}	829199368578754389926680	c_{36}	−665061729026184338578804430702
c_9	−5099785408792	c_{23}	−6126769070546369565192562	c_{37}	220816861442951387117375001273
c_{10}	121896264927527	c_{24}	−12675057918465999444140983	c_{38}	1087119449071571225723936093483
c_{11}	120570969254704	c_{25}	89040222438154792638256214	c_{39}	923955449677416630892566143 40
c_{12}	−13794186343238158	c_{26}	155697045700341519390162807	c_{40}	−1035137643751063257956261732440
c_{13}	30864849044030540	c_{27}	−931131854251471127467416134	c_{41}	−391173012111901519069217648928
c_{14}	988191969700359264	c_{28}	−1530130532385016900757918014	c_{42}	504794509042457799652653015584

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 55, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $55+4$ terms removed returns order 55 as well, so $\deg q_{\min} = 55 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 12 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $55+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 55 that this sequence satisfies — and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A235845.

[2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.